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Simulated responses from large language models are weak proxies for human economic preferences across 74 countries

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Large-scale surveys are central to investigating economic preferences but are costly and slow. Can large language model (LLM)-simulated responses substitute for human survey data? We conducted a virtual Global Preference Survey, simulating six experimentally validated economic preferences for 67,201 matched human respondents across 74 countries. Across 20 LLM configurations with diverse prompts and robustness checks, we found systematic but modest individual-level alignment (87.5% of Spearman correlations have 95% confidence intervals (CIs) above zero), with LLMs recovering the direction of associations in 77.8% (95% CI, (71.6%, 83.1%); CI-based consistency ratio) of gender, age and cognitive-ability dimensions, but only 48.1% (95% CI, (45.1%, 51.2%)) of cultural dimensions. Ablation and error decompositions suggest weaker alignment when removing demographic information than when removing country labels, and in many models alignment is weaker in economically undeveloped and less digitally connected countries. LLM-simulated responses may recover some individual- and group-level patterns but are not yet reliable proxies for cross-national representative surveys in macro-level inference or policy evaluation.

Large-scale surveys are an important tool in economics. By directly eliciting latent preferences, they complement revealed-preference approaches and support both causal and descriptive inference at the population scale[1,2]. However, implementing such surveys is expensive and slow. Researchers need to construct sampling frames, train enumerators, provide respondent incentives and coordinate complex field operations. Recent advances in large language models (LLMs) have prompted an intriguing alternative: using LLMs to generate responses for surveys or experiments[3–10]. This approach can lower costs and accelerate iterations without human recruitment.

Despite this promise, concerns remain about using LLMs' simulated responses to replace human data, given inherent algorithm biases such as identity flattening[11]; systematic skew towards Western, educated, industrialized, rich and democratic populations[12–16]; overestimated statistical power[7,17]; and other potential risks[18–21]. Prior work has focused largely on political opinions[6,17,22–26] and on social and moral attitudes[12–14,27–29], while paying much less attention to economic preferences, even though these preferences play an important role in linking individual behaviour to market outcomes and social welfare, and constitute one of the most compelling use cases for LLMs in efforts to improve individual and societal well-being.

Preference heterogeneity is a building block of modern economic theory: it helps explain individual decisions[30–32], socio-economic status[33,34] and macroeconomic outcomes[35–37], and it is indispensable for policy design and evaluation[38–40]. Understanding group-level heterogeneity in preferences is especially important in development economics, which seeks to capture how policies affect different populations rather than only the average effect. If LLMs can be shown to

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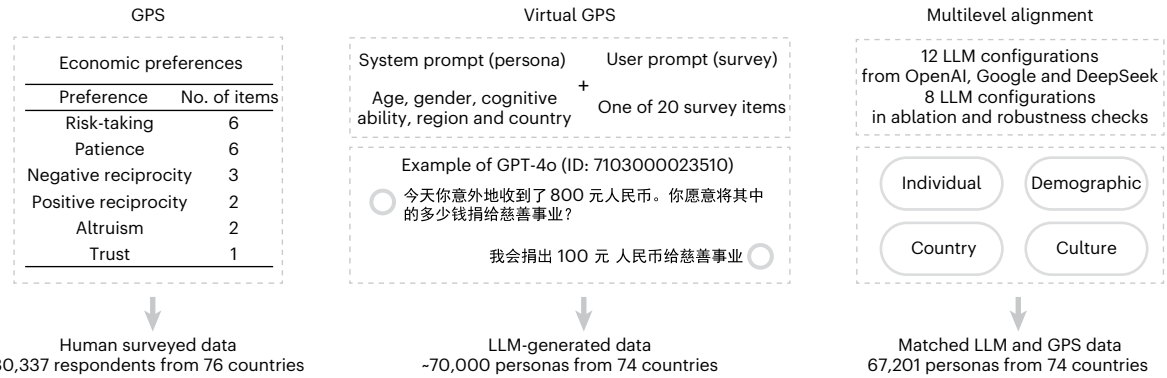


Fig. 1 | Study design. Following the GPS, we focused on six economic preferences: risk-taking, patience, negative reciprocity, positive reciprocity, altruism and trust. For each item, respondent demographics and locations were encoded in a system prompt, the GPS question text was provided as a user prompt to LLMs and the model's output was recorded as its predicted response. In the prompts, the language and incentives were localized, as in this example, where we used Chinese and RMB. (The English translation of this survey item is 'Today you unexpectedly received 800 RMB. How much of this amount would you donate to a good cause?' The translation of the LLM's answer is 'I would donate 100 RMB to charity.') We assessed alignment between LLM-predicted and GPS-surveyed (human) preferences at the individual, demographic, country and cultural levels. Twenty LLM configurations (nine local-language conditions plus three English-only conditions, five ablation conditions and three robustness checks) from OpenAI, Google and DeepSeek generated predicted responses for GPS respondents in 74 countries.

successfully predict human economic preferences, they may be used to generate counterfactual data for policy analysis, potentially alleviating the fundamental problem of causal inference—namely, that the counterfactual is unobserved[41]. However, the credibility of such applications depends on the models' ability to reproduce well-documented patterns of heterogeneity. This challenge is not trivial, given pronounced demographic variation[42–46] and culturally specific patterns[42,47–50] in economic preferences.

We used the Global Preference Survey (GPS) to perform a global, multilevel audit of LLM-generated synthetic data across individuals, demographic groups, countries and cultures[42]. The original GPS includes responses from more than 80,000 individuals across 76 countries that have been validated against incentivized experimental tasks (see Methods for details). It thus offers a rich human benchmark for our tests. The GPS also categorizes economic preferences into six distinct domains—risk-taking, patience, positive and negative reciprocity, altruism and trust—that span the core dimensions of economic decision-making[42,51]. Risk-taking captures attitudes towards uncertainty, while patience governs intertemporal trade-offs, predicting portfolio choice, entrepreneurial entry, educational attainment and saving behaviour[44,52]. Positive and negative reciprocity capture the inclination to reward kindness and punish unkindness, underpinning contract enforcement and workplace cooperation[53]. Altruism drives charitable giving and prosocial behaviour[54], and trust is a key correlate of financial development and economic growth[55]. Together, these preferences shape aggregate market outcomes (for example, savings rates, charitable giving and contract enforcement) and the effectiveness of policy interventions ranging from retirement-savings nudges to public-good provision. Validating LLM-generated responses against this six-dimensional benchmark therefore provides a test of whether LLMs can support both behavioural science and applied policy research.

Specifically, we prompted LLMs using a one-to-one design to build digital-twin panels of the GPS. That is, for each human respondent, we constructed a matched persona and prompted each LLM to answer the same questionnaire. A second key feature is that we used localized-language prompts rather than English-only prompts, to better reflect local context[56,57]. We also included English-only prompting conditions on a subset of models as a robustness check. Figure 1 illustrates our study design.

Overall, our 'virtual GPS' tested 12 LLM configurations, comprising nine local-language conditions (GPT-3.5-turbo, GPT-4o-mini, GPT-4o, GPT-4.1-mini, GPT-5.4-mini, GPT-5.4, Gemini-2.0-Flash, Gemini-2.5-Flash and DeepSeek-V3) and three English-only conditions (GPT-4o-mini, GPT-5.4-mini and GPT-5.4; robustness check for prompt language), with each configuration completing the original GPS module for about 70,000 personas. We conducted eight follow-up experiments using GPT-5.4-mini: five prompt-component ablations, one paraphrasing variant, one response-constraint variant and one high-temperature variant. Each persona embedded age, gender, cognitive ability and country/region, and prompts were delivered in the respondents' original survey language (except English-only configurations; see Methods and Supplementary Information section A for details). In total, this protocol yielded about 27.8 million LLM-generated answers and roughly 1.39 million personas (~70,000 personas × 20 configurations) that are directly comparable to the human GPS benchmark (see Supplementary Tables A.1 and A.2 for dataset descriptions). Together with the GPS benchmark, our design allows for fine-grained comparison of LLM-generated responses with human data at multiple levels.

Our study addresses several questions that are both theoretically and practically relevant. First, we provide a cross-country, multilevel audit of the validity of LLM-generated responses using a benchmark that is theoretically grounded, experimentally validated and widely used in cross-country economic research[37,58–63]. Second, we clarify the conditions under which LLM-generated responses are informative, by assessing whether synthetic preference data can plausibly support the study of group-level heterogeneity across demographic and national contexts. Third, we relate cross-country variation in LLM performance to macro-level characteristics such as economic development and digital infrastructure, providing a framework for diagnosing where LLM-generated responses are more versus less reliable and for thinking about how more inclusive models might be developed. Fourth, we present a protocol based on persona-level, language-matched prompting that can be used to track model updates over time, compare architectures and integrate LLM audits into ongoing economic and policy research. Taken together, this study evaluates the conditions under which LLMs can be used responsibly for policy-relevant measurement and analysis in global economic research.

Results

LLM-predicted preferences align positively but weakly with human preferences

We first examined individual-level alignment by computing Spearman correlations between LLM-predicted and GPS-surveyed preferences (see Methods for details on the construction of individual-level

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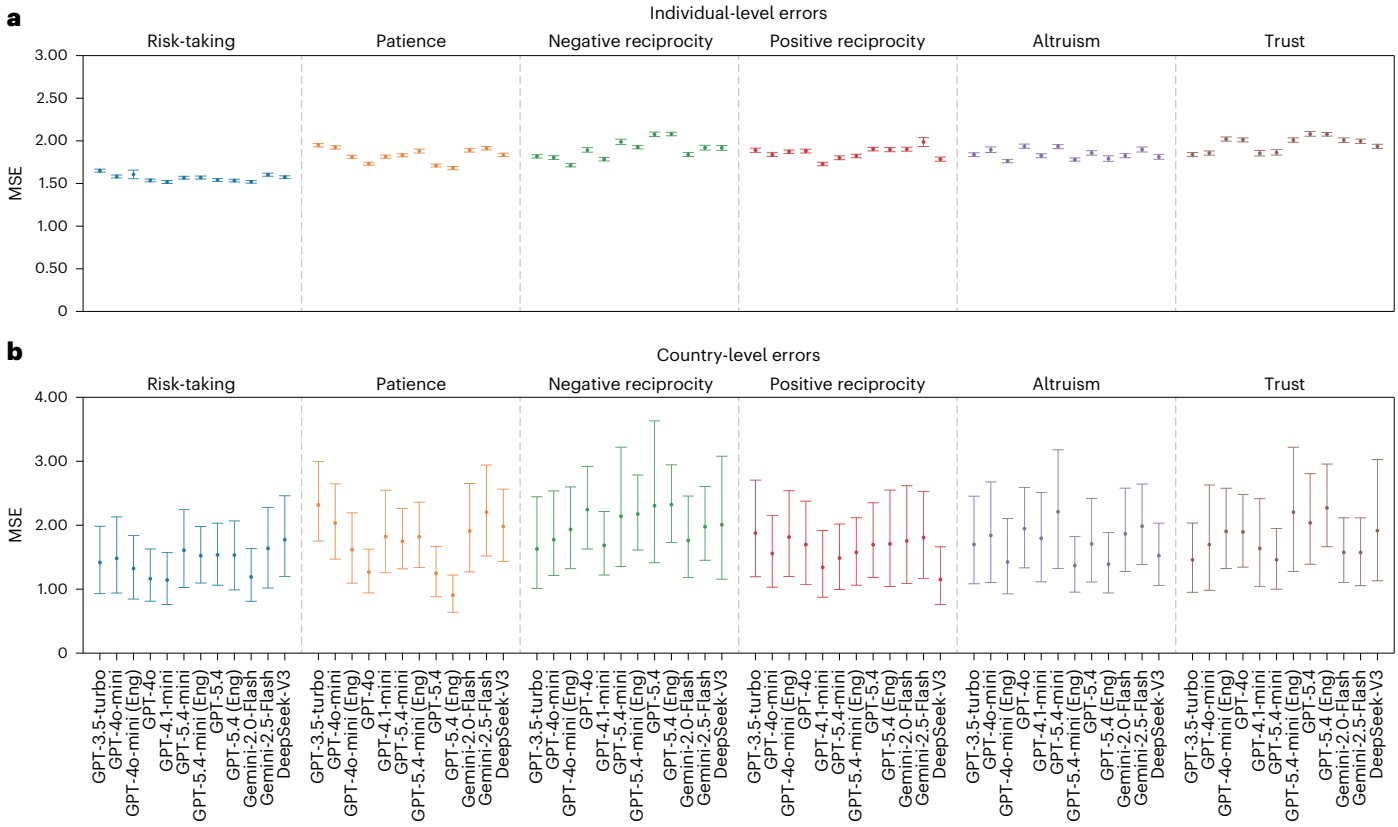


Fig. 2 | MSEs between LLM-predicted and GPS-surveyed preferences. **a**, Individual-level MSEs. Each point represents the MSE between LLM-predicted and GPS-surveyed preferences, pooling all individual observations across 74 countries (n = 67,201). **b**, Country-level MSEs. Each point represents the MSE between country-level average LLM-predicted and GPS-surveyed preferences (n = 74). Country-level averages are weighted by the GPS ex-post sampling weights. Before the MSEs were computed, both the individual- and country-level preference indices were normalized to mean 0 and standard deviation 1. Error bars indicate 95% bootstrapped CIs in both panels.

preferences), pooling all responses from 74 countries. Sixty-three of 72 (87.5%) correlations (6 preference domains × 12 LLM configurations) are significantly positive (that is, the 95% bootstrapped confidence interval (CI) excludes zero; 63 of 72 based on Benjamini–Hochberg false discovery rate (FDR)-adjusted P values, $P < 0.05$; Supplementary Table B.4), indicating that LLM-predicted preferences are directionally aligned with GPS-surveyed preferences (Supplementary Fig. B.1). Repeating the analysis within each country yielded a similarly high proportion. On average, 2,603 of 5,327 correlations are significantly positive after Benjamini–Hochberg FDR correction (Supplementary Table B.3, Supplementary Figs. B.2–B.13 and Supplementary Data 1).

Next, we quantified the discrepancy between LLM-predicted and GPS-surveyed preferences using mean squared error (MSE), where lower values indicate closer alignment. Figure 2a shows that the mean of 72 MSEs (6 preference domains × 12 LLM conditions) is 1.827 (95% bootstrapped CI, (1.793, 1.858); the full MSE estimates are reported in Supplementary Table B.4). Because the individual-level preference indices are normalized to mean 0 and standard deviation 1 prior to MSE computation (Methods), the MSEs are nearly twice the variance of the human benchmark, indicating that individual-level alignment between LLM-predicted and GPS-surveyed preferences is fairly weak.

Across preference domains, the average MSEs across all models are 1.567 for risk-taking, 1.832 for patience, 1.897 for negative reciprocity, 1.859 for positive reciprocity, 1.847 for altruism and 1.963 for trust. Turning to model comparisons, we found no evidence that newer models such as GPT-5.4 and GPT-5.4-mini outperform earlier OpenAI models (that is, GPT-3.5-turbo, GPT-4o-mini, GPT-4.1-mini and GPT-4o; two-sided Welch's t-test; $t_{41.32} = 0.851$; Benjamini–Hochberg FDR-corrected $P = 0.726$; Cohen's $d = 0.241$; 95% CI, (−0.05, 0.12); Bayes Factor (BF_{10}) = 0.372) or that Gemini-2.5-Flash outperforms Gemini-2.0-Flash (two-sided Welch's t-test; $t_{9.83} = 0.612$; Benjamini–Hochberg FDR-corrected $P = 0.726$; Cohen's $d = 0.353$; 95% CI, (−0.15, 0.26); Bayes Factor (BF_{10}) = 0.527).

We next examined country-level alignment by aggregating GPS-surveyed and LLM-predicted preferences to country means using the GPS ex-post sampling weights (see Methods for details on the construction of country-level preferences). We first computed split-half internal consistency for country-level averages. The split-half internal consistency of LLM-predicted preferences is comparable to, and slightly higher than, that of the GPS-surveyed data ($M_{GPS} = 0.984$ versus $M_{LLMs} = 0.992$; two-sided Welch's t-test; $t_{8.00} = 4.353$; $P = 0.002$; Cohen's $d = 1.142$; 95% CI, (0.00, 0.01); see Supplementary Information section B.3 for details). This supports the feasibility of constructing country-level average preference indices from LLM-predicted preferences. Qualitatively, of the 72 country-level correlations between LLM-predicted and GPS-surveyed preferences (6 preference domains × 12 LLM configurations, each computed across the 74 country means), 58 are positive, but only 22 have 95% bootstrapped CIs that exclude zero (16 of 72 based on Benjamini–Hochberg FDR-adjusted P values, $P < 0.05$; Supplementary Fig. B.1 and Supplementary Table B.5). The country-level MSEs are reported in Fig. 2b. The mean country-level MSE across preferences is 1.722, slightly lower than its individual-level counterpart ($M_{country} = 1.722$ versus $M_{individual} = 1.827$; two-sided paired t-test; $t_{71} = -3.744$; $P < 0.001$; Cohen's $d = -0.425$; 95% CI, (−0.16, −0.05); see Supplementary Table B.5 for all country-level MSE estimates), probably because aggregating responses within countries removes a lot of the noise in individual-level responses. It is worth

noting that despite this, the MSEs remain large, compared with their standard deviations.

Finally, to interpret the above results, we calculated the proportion of variance in GPS-surveyed preferences explained by LLM-predicted preferences. We estimated this by running ordinary least squares (OLS) regressions of GPS-surveyed preferences on LLM-predicted preferences at both the individual and country levels and calculating the resulting R^2. This analysis shows that LLM-predicted preferences explain only 1.3% of the individual-level variance and 4.2% of the country-level variance in human preferences (two-sided paired t-test; $t_{71} = 5.196$; $P < 0.001$; Cohen's $d = 0.722$; 95% CI, (0.02, 0.04); Supplementary Tables B.1 and B.2). Again, the higher R^2 for country-level predictions probably reflects the fact that aggregating responses within countries removes a lot of noise compared with the individual-level data. Nonetheless, both sets of R^2 values indicate that LLM-predicted preferences exhibit only weak positive alignment with the human benchmark.

To assess whether these patterns are model-specific or reflect a general regularity, we next compared LLM-predicted preferences across models. We first examined between-model agreement using Spearman correlations. At both the individual and country levels, LLM-predicted preferences are predominantly positively correlated across models (380 of 396 are significantly positive after Benjamini–Hochberg FDR correction, ranging from 0.676 for risk-taking to 0.092 for trust across preference domains, with a mean correlation of 0.346; 197 of 396 country-level correlations are significantly positive after Benjamini–Hochberg FDR correction, ranging from 0.363 for altruism to 0.201 for trust, with a mean correlation of 0.262; see Supplementary Figs. C.1 and C.2; all between-model correlations with 95% CIs and FDR-adjusted P values are reported in Supplementary Data 2).

We then quantified cross-model agreement more directly using between-model MSEs. Figure 3 shows the individual-level between-model MSE matrices for each pair of LLMs. The mean individual-level between-model MSE across preferences is 1.377 (risk-taking, 0.770; patience, 1.049; negative reciprocity, 1.679; positive reciprocity, 1.579; altruism, 1.329; trust, 1.856), lower than the corresponding LLM–GPS MSE of 1.827 (two-sided Welch's t-test; $t_{313.31} = -16.569$; $P < 0.001$; Cohen's $d = -1.158$; 95% CI, (−0.50, −0.40); see Supplementary Table C.2 for preference-level comparisons). The mean country-level between-model MSE is 1.491 (risk-taking, 1.423; patience, 1.500; negative reciprocity, 1.628; positive reciprocity, 1.485; altruism, 1.199; trust, 1.709), lower than the country-level LLM–GPS MSE of 1.722 (two-sided Welch's t-test; $t_{115.62} = -5.468$; $P < 0.001$; Cohen's $d = -0.600$; 95% CI, (−0.32, −0.15); see Supplementary Fig. C.3 and Supplementary Table C.2 for the country-level matrix and preference-level comparisons).

These results suggest that, on average, LLM-generated preference estimates are broadly similar across models, particularly at the individual level, yet this convergence aligns only weakly with the GPS-surveyed preferences, indicating that current LLMs share a common output regularity that systematically departs from the human benchmark.

Finally, we note that alignment is not uniform across preference domains. Descriptive evidence shows that individual- and country-level MSEs in Fig. 2 are consistently low for risk-taking and high for trust (mean individual-level MSE, 1.567 versus 1.963; country-level, 1.444 versus 1.802), mirroring the between-model MSE patterns (individual-level in Fig. 3, country-level in Supplementary Fig. C.3; mean individual-level MSE, 0.770 versus 1.856; country-level, 1.423 versus 1.709). This convergence between human–model alignment and between-model consistency suggests that LLM-simulated responses may recover some preference domains more readily than others, with trust showing both large deviations from human data and low between-model agreement across architectures.

LLM-predicted preferences recover associations with age, gender and cognitive ability but are weak proxies for cultural differences

We next investigated whether LLM-predicted preferences reproduce the demographic gradients with respect to gender, age and cognitive ability observed in the GPS data. For each LLM and preference domain, we computed effect sizes (Cohen's d) and their 95% CIs for gender (male versus female), age (younger versus older) and cognitive ability (high versus low) (see Methods for details). We then plotted each LLM-based effect size against its GPS counterpart and classified the resulting (GPS, LLM) pair by quadrant, as reported in Fig. 4 (see Supplementary Data 3 for details).

We assessed demographic alignment using a 95%-CI-based consistency criterion (see Methods for details) and report the resulting consistency ratios with their 95% CIs in the main text. As a robustness check, Supplementary Information section D.2 reports a sign-based match ratio, defined as the proportion of GPS–LLM effect-size pairs with the same sign, irrespective of statistical significance. The overall demographic-level consistency ratio is 77.8% (two-sided binomial test; proportion, 168/216; $P < 0.001$; 95% CI, (71.6%, 83.1%); gender, 81.9%; age, 66.7%; cognitive ability, 84.7%). These relatively high consistency ratios indicate that LLM-predicted preferences reproduce most of the demographic patterns documented in the GPS benchmark, especially in domains such as risk-taking (100%), patience (100%) and negative reciprocity (88.9%), but less so for positive reciprocity (38.9%).

Demographic differences, however, capture only one source of variation in economic preferences. A large literature shows that preferences also vary systematically across cultural contexts[48,49,64–67]. We therefore examined whether LLM-predicted preferences recover cross-cultural differences along cultural dimensions from the World Values Survey (operationalized via the Inglehart–Welzel cultural map), Hofstede's cultural dimensions and Schwartz's cultural value orientations[68–70]. For each dimension, we applied a median split to classify countries as high or low, inherited the classification at the individual level and computed Cohen's d between high and low groups, paralleling the demographic analysis (Methods).

Figure 5 reports cultural-level LLM-based and GPS-based effect sizes. The overall cultural-level consistency ratio is 48.1% (two-sided binomial test; proportion, 520/1,080; $P = 0.235$; 95% CI, (45.1%, 51.2%)), substantially lower than the demographic-level consistency ratio (chi-squared test; $\chi^2_1 = 62.27$; $P < 0.001$; Cramer's $V = 0.22$; 95% CI, (−35.5%, −22.9%); see Supplementary Table D.1 for details). Across preferences, risk-taking achieves the highest consistency ratio (67.2%). Across cultural dimensions, World Values Survey traditional values ranks the highest (62.5%), yet remains below the lowest demographic gradient (age, 66.7%). Performance is particularly weak for some widely studied cultural dimensions. For example, according to the consistency-ratio definition, only three models (out of 12) recover the relationship between Hofstede's long-term orientation and patience in GPS, and likewise only one model recovers the relationship between Hofstede's uncertainty avoidance and risk-taking in GPS (see Supplementary Table D.1 for details).

Both CI-based consistency ratio and sign-based match ratio metrics yield the same qualitative conclusion: alignment at the cultural level is markedly weaker than at the demographic level. Moreover, Supplementary Information section D.3 shows that effect sizes for demographic and cultural gradients are positively correlated across LLMs (see also Supplementary Table D.6), indicating that LLMs tend to make similar directional predictions even when they jointly diverge from the GPS.

Decomposing the sources of LLM–GPS misalignment

Why are LLM-predicted responses only weakly aligned with GPS-surveyed preferences, and why do they recover cultural patterns so poorly? One possibility is that country information in the prompts

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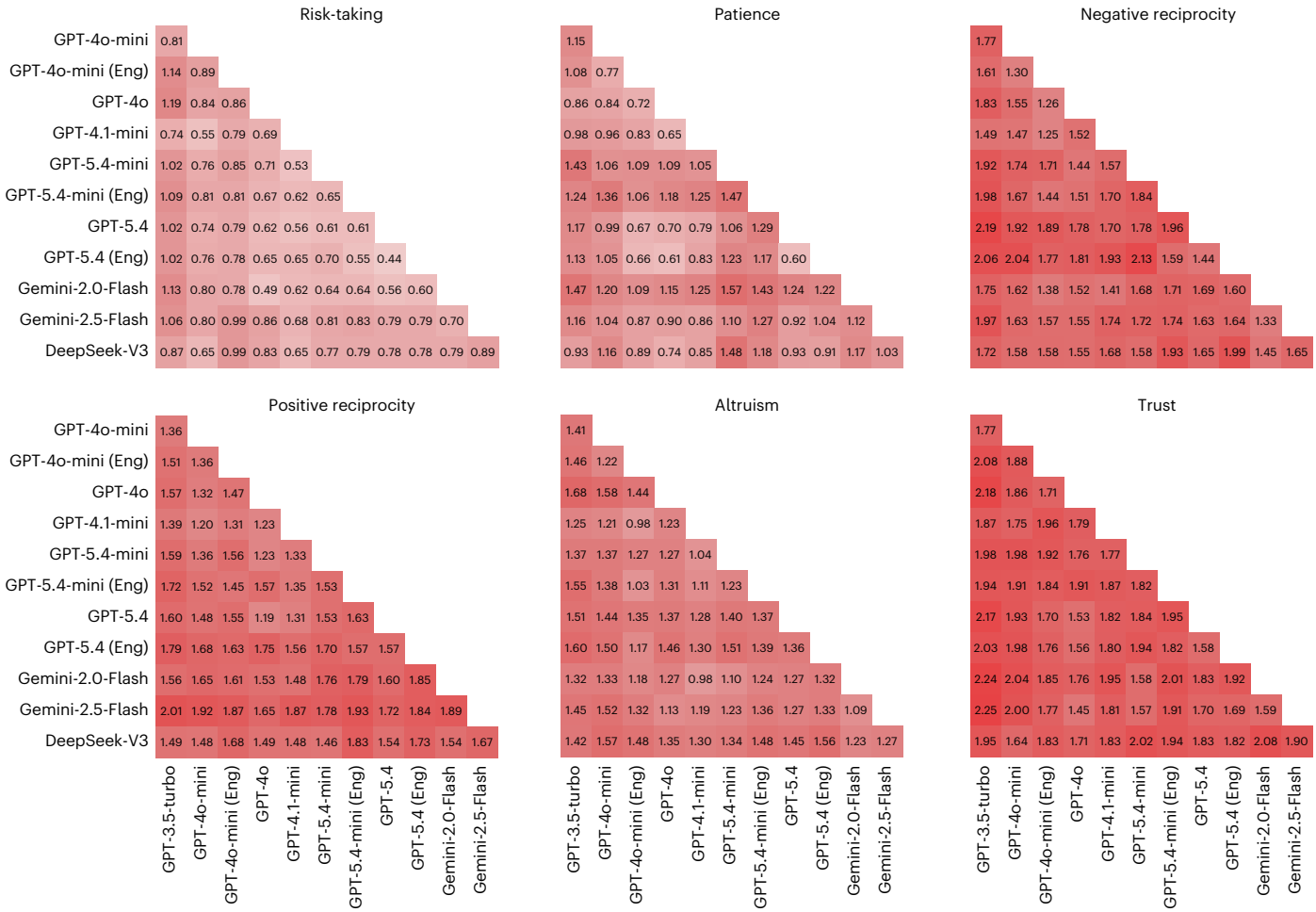


Fig. 3 | Individual-level between-model MSEs of LLM-predicted preferences. Each cell of the 12 × 12 matrix reports the MSE between two LLM configurations, computed by pooling individuals across all 74 countries. Individual-level preference indices were standardized to mean 0 and standard deviation 1 prior to MSE computation. Colour intensity encodes MSE magnitude, with lower values indicating greater consistency between models. The corresponding country-level between-model MSE matrix is reported in Supplementary Fig. C.3.

plays only a limited role, with models failing to condition meaningfully on it. To probe this, we regressed GPS-surveyed preferences on LLM-predicted preferences and progressively added covariates: a baseline with no additional covariates, then country indicators, demographic characteristics and both sets jointly. If the coefficient of LLM-predicted preferences shrinks after adding a given set of controls, this suggests that the corresponding variables account for part of the observed alignment (Methods).

Figure 6a shows that adding country indicators leaves the regression coefficients essentially unchanged, whereas adding demographic controls substantially attenuates them. A linear mixed-effects regression across model × preference combinations confirms this pattern (relative to the baseline: $\beta_{\text{Country}} = 0.000$; $P = 0.961$; 95% CI, (−0.014, 0.015); $\beta_{\text{Demo}} = -0.066$; $P < 0.001$; 95% CI, (−0.081, −0.052); $\beta_{\text{Demo+Country}} = -0.073$; $P < 0.001$; 95% CI, (−0.088, −0.059); Supplementary Information section E.1 and Supplementary Data 4). These results indicate that demographic information in the prompts accounts for more of the individual-level alignment than country labels do.

To investigate the sources of the weak alignment at the country and cultural levels, we constructed a country-level measure, the inconsistency index, which is defined as the GPS sampling-weighted within-country mean of individual-level MSEs between LLM-predicted and GPS-surveyed preferences (see 'Alignment metrics' in Methods). We related these indices to four country characteristics that plausibly track LLMs' training conditions. Higher-income countries may attract greater optimization effort from LLM developers, as they tend to host larger user bases[71,72], increasing the commercial returns to improving model performance for these contexts. We used log gross domestic product (GDP) per capita and the Information and Communications Technology Development Index (IDI) as proxies for the amount of compute used for training, and log population and log number of internet users as proxies for corpus size[73]. Figure 6b reports the Spearman correlations between these characteristics and the country-level inconsistency index (all correlations with 95% CIs and FDR-adjusted *P* values are reported in Supplementary Data 5).

We observed significantly negative correlations between the inconsistency index and GDP or IDI (36/72 (50.0%) for GDP and 31/72 (43.1%) for IDI are significantly negative after Benjamini–Hochberg FDR correction). In contrast, population size and the number of internet users show little systematic relation to the inconsistency index (none of them are significantly negative after Benjamini–Hochberg FDR correction). Higher levels of economic development and digital infrastructure are thus associated with smaller prediction errors in LLM-predicted preferences in many models, but the same pattern was not observed among proxies for corpus size. Two robustness checks yielded similar conclusions: substituting a country-level alignment measure (the consistency index; Supplementary Information section E.2) and substituting an alternative inconsistency index defined as the squared difference between sampling-weighted country-level averages (Supplementary Information section E.3).

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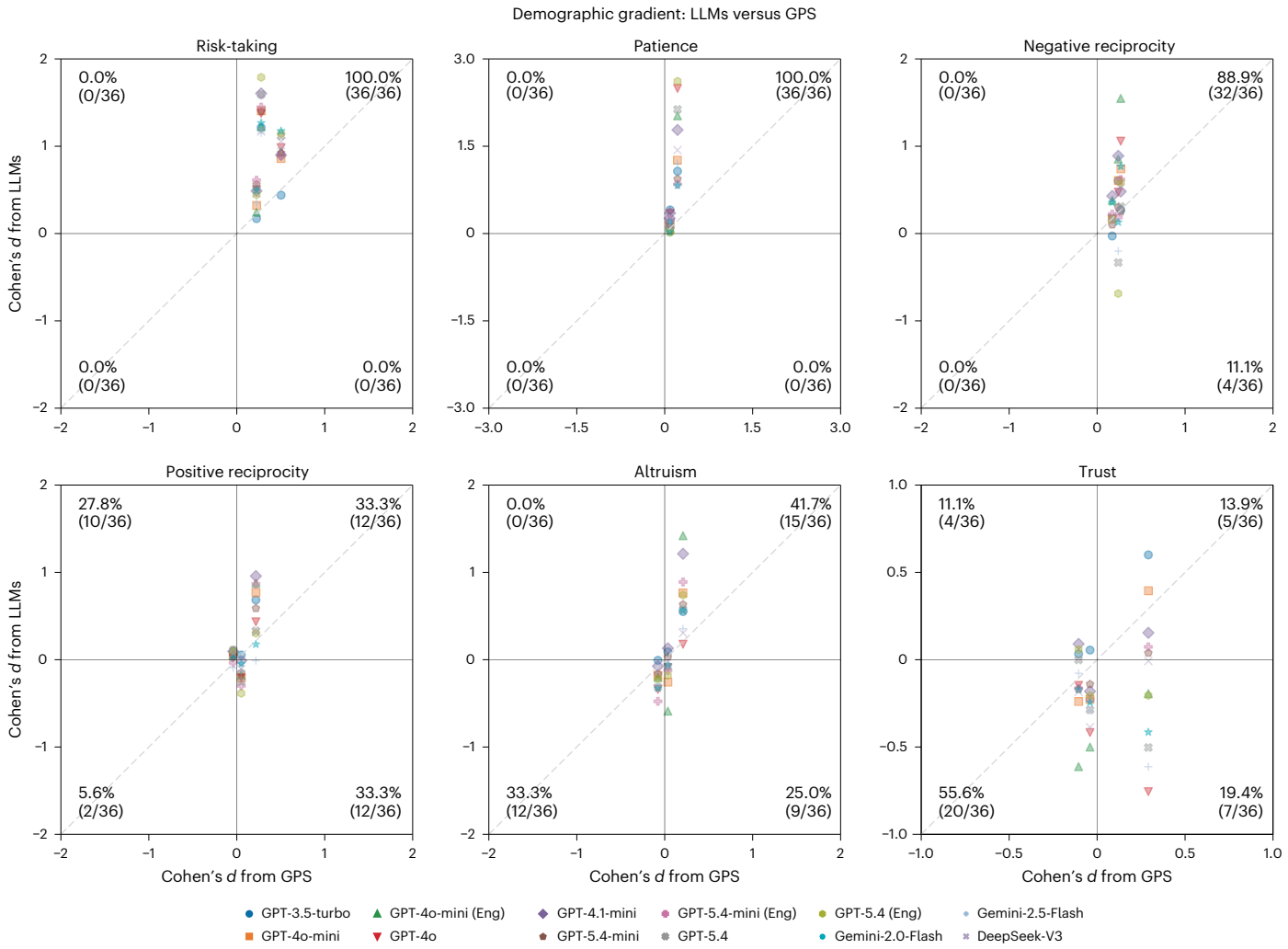


Fig. 4 | Demographic gradients in GPS-surveyed and LLM-predicted preferences. Each panel corresponds to one of the six preference domains. In each panel, the *x* axis shows effect sizes (Cohen's *d*) estimated from GPS-surveyed preferences, and the *y* axis shows the corresponding effect sizes estimated from LLM-predicted preferences. Demographic variables are encoded as binary indicators for gender (male versus female), age (<60 versus ≥60) and cognitive ability (≥5 versus <5). Each point represents one (LLM × demographic dimension) pair; with 12 LLM configurations and three demographic dimensions (gender, age and cognitive ability), each panel contains 36 points. Positive effect sizes indicate higher preference levels for males (versus females), younger (versus older) and higher cognitive ability (versus lower). The dashed 45° line denotes exact consistency between LLM- and GPS-derived effect sizes in both sign and magnitude. The percentage in each quadrant reports the share of the 36 (LLM × demographic dimension) pairs falling in that quadrant, and Quadrants I (upper right) and III (lower left) jointly indicate sign-consistent alignment with GPS (that is, the match ratio). Marker shape and colour denote the LLM configuration (see legend). Axis ranges differ across panels for visualization clarity and should not be compared across panels.

We further decomposed individual-level prediction errors (MSEs) by regressing them on demographic variables, country characteristics and country fixed effects (Fig. 6c and Methods; Supplementary Information section E.4 reports the results for each preference). Two patterns emerged. First, country fixed effects yield R^2 values ranging from 3% to 37% across models, though this partly reflects the high dimensionality of the fixed-effects specification. Second, among observable country characteristics, economic-related variables explain more variance in prediction errors than population-related variables, although their explanatory power remains modest (up to nearly 7%, from DeepSeek-V3).

Together, these analyses indicate that, compared with country label, demographic information in prompts accounts for more of the individual-level alignment, whereas the prediction errors that remain are concentrated along country lines and track indicators of economic development rather than corpus-size proxies. This pattern points to systematic, country-structured weaknesses in LLM-predicted preferences that prompt-level country labels alone do not remedy.

Diagnostic evidence from ablation experiments

To assess the contribution of each prompt component to alignment, we conducted a series of diagnostic prompt-component ablation experiments, removing the information contained in system prompts step by step (Methods). We compared four types of conditions (six in total): full-persona (containing gender, age, cognitive ability and country information); three partial-persona variants, each removing one demographic characteristic (gender, age or cognitive ability) while retaining the others and country; demographic-persona (all demographic characteristics, no country); and country-persona (country only, no demographic characteristics).

We first examined which type of information contributes most to the alignment. As shown in Fig. 7a, the individual-level MSE is 1.835 for demographic-persona, 1.838 for full-persona and 1.961 for country-persona (Supplementary Information section F.2). These values suggest an asymmetric contribution between the two types of prompt information. A linear model with preference-domain fixed effects confirms this pattern (relative to the country-persona

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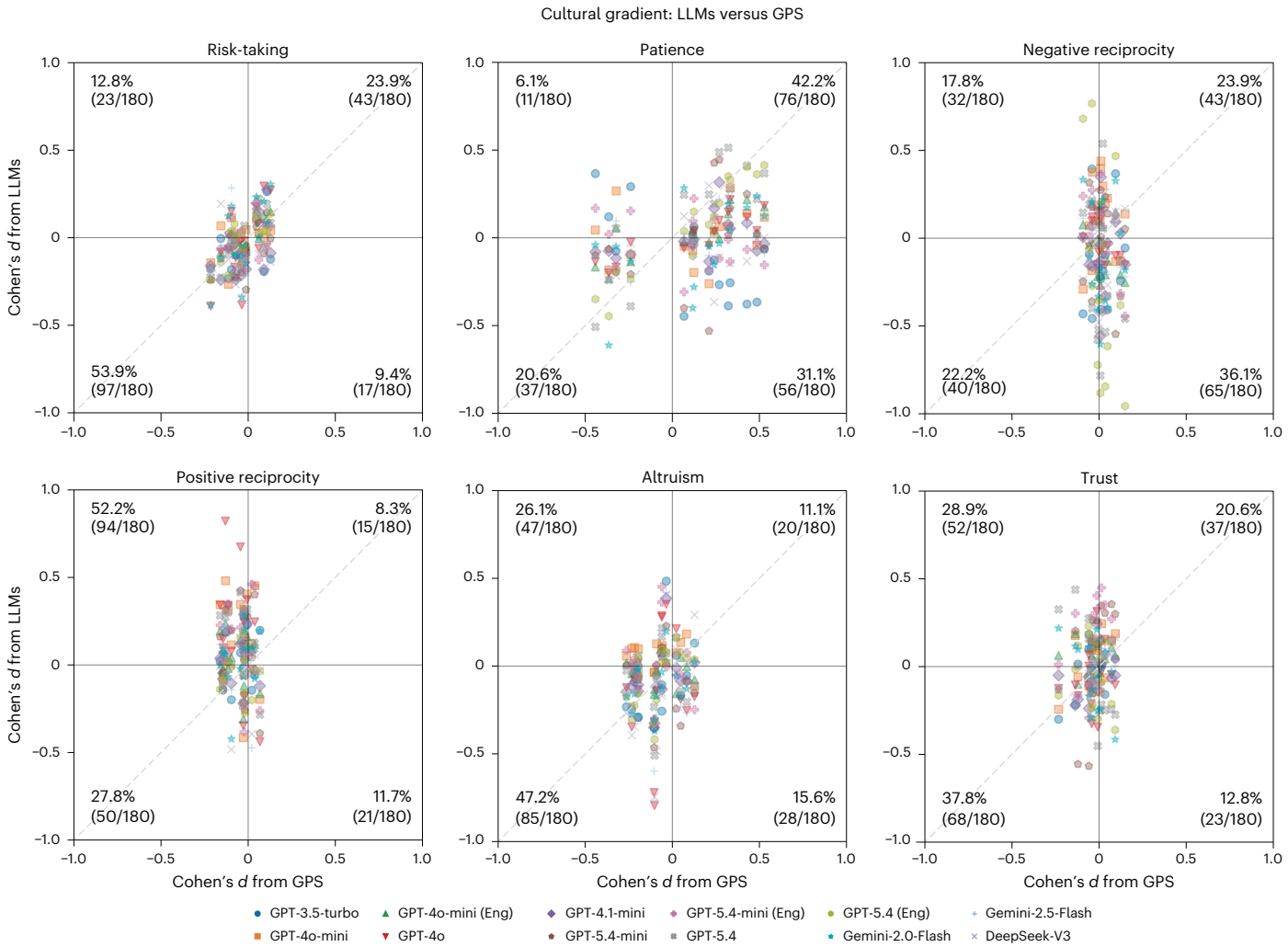


Fig. 5 | Cultural-level consistency and gradients in GPS-surveyed and LLM-predicted preferences. Each panel corresponds to one of the six preference domains. In each panel, the *x* axis shows effect sizes (Cohen's *d*) estimated from GPS-surveyed preferences, and the *y* axis shows the corresponding effect sizes estimated from LLM-predicted preferences. For each cultural dimension, countries are split at the median into low and high groups, and individual respondents inherit the classification of their country. Each point represents one (LLM × cultural dimension) pair; with 12 LLM configurations and 15 cultural dimensions, each panel contains 180 points. Positive effect sizes indicate higher preference levels in high-group countries relative to low-group countries. The dashed 45° line denotes exact consistency between LLM- and GPS-based effect sizes in both sign and magnitude. The percentage in each quadrant reports the share of the 180 (LLM × cultural dimension) pairs falling in that quadrant, and Quadrants I (upper right) and III (lower left) jointly indicate sign-consistent alignment with GPS (that is, the match ratio). Marker shape and colour denote the LLM configuration (see legend).

condition: $\beta_{\text{partial-persona}} = -0.097$; $P = 0.001$; 95% CI, (−0.149, −0.046); $\beta_{\text{demographic-persona}} = -0.126$; $P < 0.001$; 95% CI, (−0.189, −0.064); $\beta_{\text{full-persona}} = -0.123$; $P < 0.001$; 95% CI, (−0.186, −0.061); Supplementary Table F.2). The same asymmetry holds when alignment is measured by Spearman correlation (Supplementary Fig. F.1). The consistency ratio varies across the ablation conditions, but none of the changes is statistically significant (McNemar's tests, all FDR-corrected $P \geq 0.05$; see Supplementary Information section F.2).

Figure 7b further shows the effect sizes on demographic dimensions under different ablation conditions. The average effect sizes are 0.303 for full-persona, 0.303 for demographic-persona, 0.046 for partial-persona and 0.016 for country-persona (Supplementary Table F.5). A linear mixed-effects regression across model × preference × demographic-dimension combinations confirms that the recovery of demographic gradients depends on the presence of demographic information in the prompts (relative to country-persona: $\beta_{\text{full-persona}} = 0.287$; $P < 0.001$; 95% CI, (0.132, 0.442); $\beta_{\text{demographic-persona}} = 0.287$; $P < 0.001$; 95% CI, (0.132, 0.442); $\beta_{\text{partial-persona}} = 0.030$; $P = 0.708$; 95% CI, (−0.125, 0.184); Supplementary Table F.4). These results are consistent with the interpretation that models condition more effectively on demographic cues than on country labels (Supplementary Figs. F.3–F.9).

As additional robustness checks addressing the possibility that alignment reflects the memorization of GPS content in training data, we conducted a paraphrasing experiment, in which survey items were rephrased while preserving their meaning, and a series of English-only experiments, in which items were administered in English rather than in the respondents' original survey languages. If memorization primarily drove alignment, both manipulations should weaken the match with training data. However, there is no statistically significant evidence supporting this hypothesis (Supplementary Information section G).

Practical constraints of LLM-simulated responses in cross-national research

To illustrate the limitation of LLM-simulated responses in social science, we replicated published GPS-based findings using LLM-simulated responses (Supplementary Information section H). Across 35 published country-level associations (for example, patience and GDP

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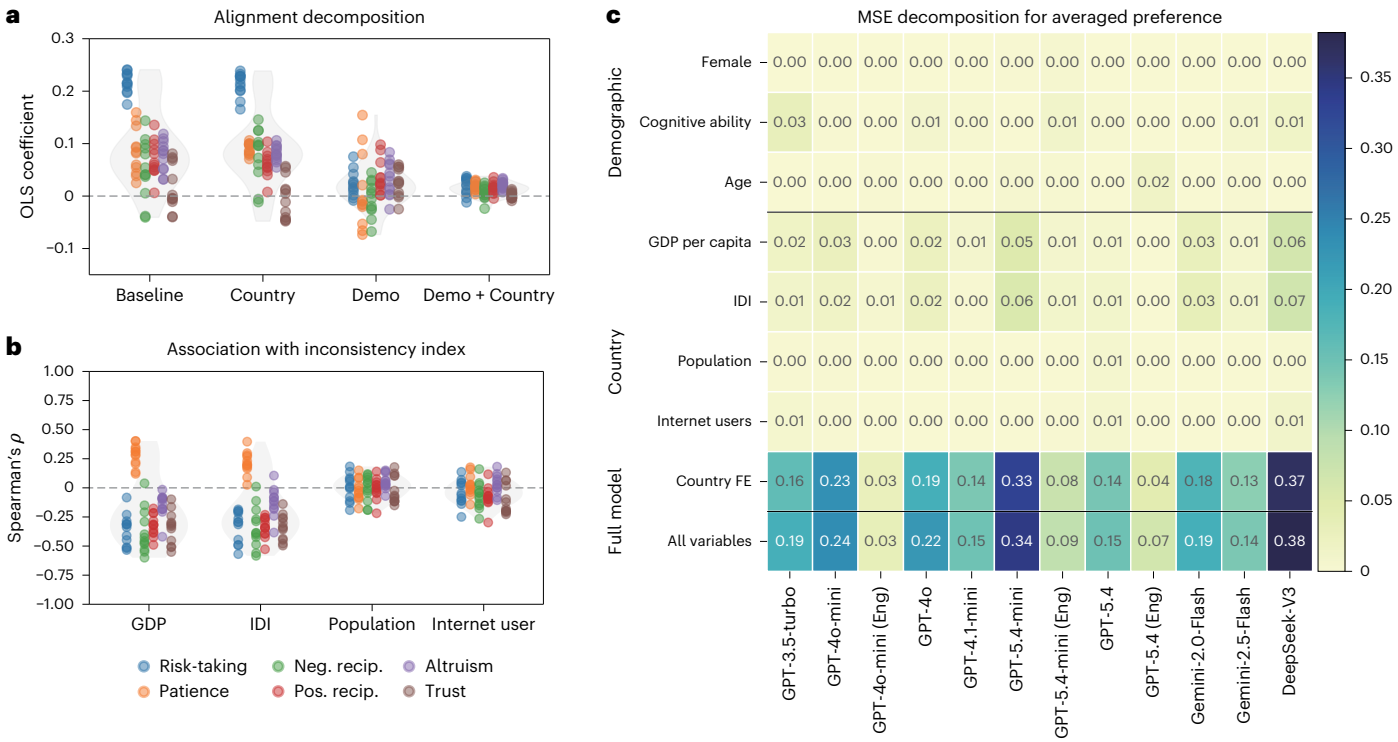


Fig. 6 | Decomposing the sources of LLM–GPS misalignment. a, Regression coefficients from models regressing GPS-surveyed preferences on LLM-predicted preferences. Each point represents an OLS coefficient estimated on pooled individual-level data for a given model × preference combination (12 LLMs × 6 preferences). The four categories on the *x* axis correspond to four alternative regression specifications: baseline (no controls), with country indicators, with demographic characteristics and with both sets jointly. **b**, Spearman correlations between the inconsistency index (the GPS sampling-weighted within-country mean of individual-level MSEs) and four country characteristics: log GDP per capita, IDI, log population and log number of internet users. Each point represents a Spearman correlation coefficient computed across countries with non-missing data (GDP, $n = 73$; IDI, $n = 72$; population, $n = 74$; internet users, $n = 68$). Each characteristic shows 72 points (12 LLMs × 6 preferences). **c**, Error decomposition of LLM-predicted preferences. For each LLM (column), separate OLS regressions were estimated with the average individual-level MSE across six preferences as the dependent variable and each variable category (row) as the independent variable. Cell values report the R^2 of each single-predictor regression. FE, fixed effects.

per capita[42]; gender gaps in preferences and gender equality[74]), the best-performing LLM reproduces only 16 (45.7%) with the same direction and statistical significance as the original estimates, using the 10% significance criterion adopted in the original studies (12 of 29 at the 5% level; Supplementary Table H.1). In contrast, the models reproduce 50–78% of 18 published individual-level demographic associations under the original 10% criterion (the range is unchanged under the 5% criterion; Supplementary Table H.2).

Beyond the replication results, we identified two additional practical constraints. One is gender bias (Supplementary Information section I). Across countries, women's prosocial preferences are systematically overestimated in LLM-predicted data relative to the GPS benchmark (altruism, 12 of 12 models; trust, 11 of 12 models; Supplementary Table I.1), and the resulting gender-gap misalignment is larger in countries with stronger gender stereotypes (Supplementary Fig. I.1). This pattern suggests that model outputs are sensitive not only to demographic-persona inputs but also to country-level gender norms embedded in the models' training corpora. The other constraint is limited response diversity. Compared with human responses in the GPS, LLM-generated responses are more concentrated[17,75], and this concentration is itself patterned by country-level development (Supplementary Information section J). Though part of this homogenization probably reflects our decision to set the temperature to zero for reproducibility, increasing the temperature to 1 only partially restores response diversity, which still remains substantially below human levels (Supplementary Fig. J.3). These limitations constrain applications of LLM-simulated responses in research, policy design and applied practice.

Discussion

How to use LLMs responsibly in empirical research is a pressing question. We have contributed to this literature by evaluating the feasibility of using LLMs to generate responses in a cross-country survey setting. Specifically, we compared LLM-generated responses with human data from the GPS across multiple levels, from individuals to demographic groups, countries and cultures, and thereby clarified the boundary conditions of their scientific utility.

Several findings stand out. LLMs show weak but systematic alignment with human data. They also reproduce a substantial share of human demographic gradients in gender, age and cognitive ability, but effect sizes may not always align. Importantly, LLM-generated responses recover cultural patterns far less reliably than demographic patterns. Moreover, prediction errors are country-structured: country fixed effects explain a substantial share of error variance, and in many models country-level performance covaries with economic development and digital infrastructure rather than population size or internet users. Finally, ablation experiments provide diagnostic evidence that demographic information in prompts contributes more to predictive accuracy than country information. Throughout, LLMs are broadly consistent with one another, even where they collectively diverge from human data.

Our results also suggest that prediction errors often track economic development and do so more closely than population. Two non-exclusive mechanisms are consistent with our evidence and with recent work on geographical biases in LLMs[12,13]. First, even with language-matched prompts, languages and content from more developed countries may receive disproportionate emphasis in LLM

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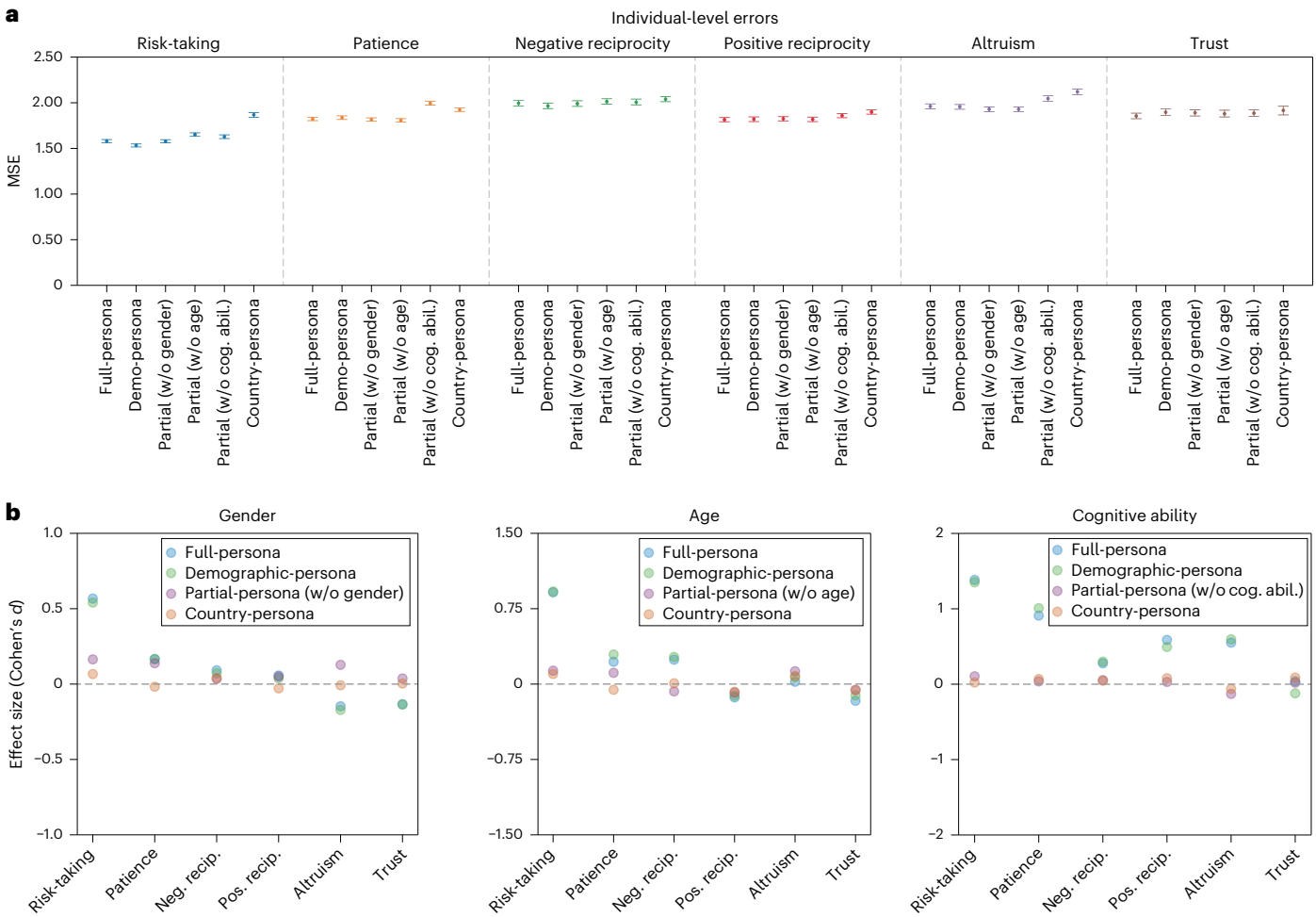


Fig. 7 | Ablation experiments isolating the contributions of country and demographic information in persona prompts. a, Individual-level errors (MSEs) across ablation conditions. Each point represents the MSE between GPS-surveyed and LLM-predicted preferences, pooling all individual observations across 74 countries (n = 65,212) for a given model × preference combination, shown for six ablation conditions: full-persona (system prompt contains gender, age, cognitive ability and country), demographic-persona (only demographic characteristics, country removed), three partial-persona conditions (full-persona with gender, age or cognitive ability removed, respectively) and country-persona (only country, all demographic characteristics removed). The error bars denote 95% bootstrapped CIs. **b**, Demographic effect sizes across ablation conditions. Each subpanel corresponds to one demographic dimension (gender, age or cognitive ability). Each point plots the Cohen's d effect size of a demographic variable (male versus female, younger versus older or higher versus lower cognitive ability) estimated from LLM-predicted preferences under each ablation condition. For each subpanel, four conditions are shown: full-persona, demographic-persona, country-persona and the matched partial-persona (for example, 'w/o gender' in the gender subpanel; see legend in each panel).The y-axis ranges differ across subpanels to accommodate effect-size magnitudes.

development and training, yielding lower errors there[76,77]. Second, developed countries are typically more digitized, providing more high-quality, structured data that can improve model performance[78].

As stated in the Universal Declaration on Cultural Diversity, "Cultural diversity is as necessary for humankind as biodiversity is for nature. In this sense, it is the common heritage of humanity and should be recognized and affirmed for the benefit of present and future generations." Our results, however, expose a practical dilemma in LLM development. Although models perform relatively poorly in lower-income countries, developers have limited commercial incentives to optimize them for these contexts. The equity risks are non-trivial. Downstream, developed countries may be able to leverage artificial intelligence more effectively in research, productivity and education, while users in developing countries receive answers that are less adapted to local contexts. Over time, if LLM performance systematically favours developed countries, cultural knowledge from developing countries may remain under-represented, encouraging homogenization and centralization of the global cultural ecosystem, eroding cross-cultural diversity[13,14] and widening digital inequity[79,80]. These risks underscore the need to align LLMs not only with universal ethical principles but also with diverse cultural, linguistic and social contexts.

Despite these constraints, our individual- and demographic-level alignment results suggest several possible scientific uses of LLM-simulated responses, although these possibilities require further empirical validation[81]. First, LLMs may be used to propose and probe latent motivational structures underlying human behaviour, particularly in Western, educated, industrialized, rich and democratic populations, complementing formal decision theory[5,21,82–85]. Second, LLMs may accelerate early-stage hypothesis generation and study-design diagnostics, helping researchers identify promising interventions or measurement strategies before turning to human samples. They might help identify when cross-cultural extrapolation may fail[86–88] and flag measurement-context mismatches[89–91]. They may also serve as a fast and low-cost environment for follow-up experiments and robustness checks[92,93] and provide a safer testbed for exploring ethically sensitive counterfactuals, such as mechanisms that increase or reduce intergroup hostility, before designing human-participant experiments. Beyond data generation, a potential

application is to use LLMs as a measurement tool[94,95], expanding social scientists' toolkit.

Beyond its implications for economic and policy analysis, our study also offers a framework for constructing multilevel alignment benchmarks for LLMs. By pairing a large-scale cross-national survey with one-to-one synthetic sampling, we obtained comparable human–model measures at the individual, demographic, country and cultural levels. The resulting metrics can be used to track how alignment evolves across successive model generations. More broadly, as LLM evaluation increasingly relies on standardized test suites for coding, reasoning and other well-defined objective tasks, there is a parallel need for benchmarks that assess predictive accuracy and cross-cultural alignment in high-impact, policy-relevant and subjective domains. Using existing large-scale cross-national surveys as behavioural benchmarks, as we do here with the GPS, provides one practical route and can lower the barrier for researchers to build more culturally diverse tests for evaluating LLMs across different contexts and subjects[96].

Several limitations qualify our conclusions. Because the GPS was conducted in 2012, portions of its content may appear in LLM training data. Successful predictions could therefore partially reflect memorization, making our alignment estimates an upper bound. As reported in the Results, however, the paraphrasing and English-only experiments provided no evidence for this account, suggesting that memorization alone is unlikely to explain the observed alignment. Additionally, secular shifts in economic preferences over the past decade could lead our analysis to underestimate alignment for current cohorts. To address both concerns simultaneously, future large-scale surveys and intervention experiments could be designed as synchronous digital-twin panels in domains where training data contamination is less likely, helping disentangle memorization from genuine reasoning.

Moreover, although we found systematic individual-level alignment, the mechanisms by which models reproduce these patterns remain opaque. At least two directions for future research are worth pursuing. The first is combining audits with targeted interventions, such as data augmentation in under-represented contexts, cross-lingual grounding and domain-specific alignment, to test whether the micro-to-macro gap can be narrowed. We note that our ablation experiments were conducted primarily with GPT-5.4-mini under a single persona-prompting design and therefore do not constitute a general test of all possible ways of encoding country or cultural context, such as richer cultural descriptions, retrieval-augmented prompts or explicit cultural calibration. A country label alone may be a weak operationalization of cultural context, and stronger operationalization might steer LLMs to behave differently. Prior work shows that cultural- and country-level alignment can be improved by prompt injection[12,80] and fine-tuning[27,97], suggesting one promising route to addressing misalignment associated with country fixed effects. The second research direction is to probe LLMs' internal representations and apply mechanistic interventions such as steering, to improve alignment[98].

Ultimately, realizing the promise of LLMs for global economic research will require sustained efforts to improve cultural and developmental coverage. Further global audits can inform constructing and validating these tools, and the framework developed here might serve as a foundation for future efforts to build more reliable and inclusive LLM-based approaches to economic and policy analysis.

Methods

This research complies with all relevant ethical regulations. Because the study only involves analyses of existing secondary data, it was determined to be exempt from ethics review by the Ethics Committee of the Neuromanagement Laboratory, Zhejiang University.

GPS

We used the GPS as the human benchmark. Conducted as part of the 2012 Gallup World Poll, the GPS includes more than 80,000 respondents from 76 countries that together account for about 90% of the world's population and global GDP. It provides one of the most comprehensive sets of globally comparable preference measures across cultures[42].

The GPS elicits six economic preference domains: risk-taking, patience, positive reciprocity, negative reciprocity, altruism and trust. Participants answer eight self-assessment items, two hypothetical choice-based quantitative items, five staircase intertemporal-choice items and five staircase lottery-choice items. The preference index is constructed by combining one abstract qualitative item with one hypothetical choice-based quantitative item into a single preference variable, except for negative reciprocity and trust (see Supplementary Table A.3 for details).

LLM-simulated survey data

To obtain LLM-predicted preferences, we conducted a large-scale one-to-one virtual survey. For each GPS respondent, we extracted demographic information (gender, age and cognitive ability) and country information and encoded them in the system prompt so that the model's responses were conditioned on that respondent's profile. The corresponding GPS question was then presented as a user prompt. To minimize stochastic variability, we set the temperature to 0, making outputs as deterministic as possible.

Following the GPS design, all payoff amounts were denominated in local currencies and calibrated accordingly. For each country, we simulated responses only in the most widely sampled survey language whenever possible. Translations from English into other languages were generated using Google Translate API. On average, we obtained about 70,000 valid observations per model, for a total of 1,390,525 observations across all conditions (see Supplementary Information section A for details, including the ablation, paraphrasing and response-constraint experiments). Our analysis sample comprised 67,201 respondents (54.7% female; mean age, 42.45 years; s.d., 17.56) across 74 countries.

Statistical analysis

For comparisons between independent samples, we used Welch's *t*-tests. Because normality cannot be guaranteed for small samples, we repeated the *t*-tests using non-parametric alternatives (Wilcoxon rank-sum tests for unpaired comparisons and Wilcoxon signed-rank tests for paired comparisons; Supplementary Information section L). For binomial tests, the null hypothesis was that the success probability equals 0.5.

Construction of individual- and country-level preference indices

To maintain comparability with the original GPS dataset, we followed its scoring and aggregation procedure for the LLM outputs. For each survey item, we first computed individual-level *z*-scores. We then constructed the six preference indices as weighted sums of item-level *z*-scores using the experimentally validated GPS weights.

To ensure comparability across models, we retained only those observations for which predictions are available from all 12 LLM configurations and restandardized the resulting preference indices. Country-level preferences were computed as sampling-weighted means of the individual indices using the ex-post sampling weights provided in the GPS.

Binary encoding of demographic and cultural variables

Age, cognitive ability and cultural dimensions were encoded as binary indicator variables. Older respondents are those aged 60 years or above. Cognitive ability in the GPS is measured on an 11-point Likert scale (0–10) on the basis of self-assessed agreement with the statement 'I am good at math'. Scores below 5 are coded as low cognitive ability and scores of 5 or above as high.

For cultural dimensions and other country-level variables, we constructed binary indicators using country-level medians. For each cultural dimension (for example, World Values Survey traditional values), countries at or below the median are coded as low, and those above the median as high. Each individual inherits the cultural classification of their country.

Alignment metrics

We quantified overall alignment between LLM-predicted and GPS-surveyed preferences using two complementary metrics. First, Spearman correlations were computed at the individual and country levels. Second, the MSE captures the average squared deviation between LLM-predicted and GPS-surveyed preferences. We report MSE at both the individual and country levels. The individual-level MSE pools all individual observations across countries, while the country-level MSE is computed between LLM-predicted and GPS-surveyed country-level averages, where country-level averages are weighted by GPS ex-post sampling weights. Before computing the MSE, both individual- and country-level preferences were normalized to mean 0 and standard deviation 1.

For country-specific alignment, we used two complementary metrics. The consistency index is a country-level Spearman correlation, computed within each country by pooling all individual respondents from that country. The inconsistency index is the GPS sampling-weighted within-country mean of individual-level MSEs. Higher consistency and lower inconsistency indicate better country-level alignment. Statistical analyses were performed using the Pingouin Python package (v.0.5.5)[99].

Consistency ratio

We used a consistency ratio to quantify how well the LLM-predicted preferences reproduce demographic and cultural gradients observed in the GPS. For each binary grouping variable (gender, age, cognitive ability or high versus low on a given cultural dimension) and each preference domain, we computed effect sizes (Cohen's *d*) for the GPS data and for each LLM. For example, the effect of cognitive ability on risk-taking is summarized by comparing the mean standardized risk-taking score of respondents with higher versus lower cognitive ability and dividing this difference by the pooled standard deviation. These effect sizes provide a unit-free measure of the average difference between the two groups, with positive values indicating higher preference levels in the reference group (for example, males, younger individuals, higher ability or high cultural dimension).

An effect is counted as consistent if either (a) the 95% CIs for both GPS-based and LLM-based effect sizes include zero (a concordant null effect) or (b) the 95% CIs for both effect sizes exclude zero and the two effect sizes have the same sign. The consistency ratio is defined as the proportion of such consistent cases across all model-preference combinations. Higher consistency ratios indicate stronger alignment between LLM-predicted and GPS-surveyed patterns of heterogeneity.

Linear mixed-effects regressions

To assess how much demographic and country information contributes to individual-level alignment, we first estimated four OLS regressions for each model × preference combination, regressing GPS-surveyed preferences on LLM-predicted preferences with (1) no covariates (baseline), (2) demographic controls only (gender, age and cognitive ability), (3) country dummies only and (4) both demographic controls and country dummies. This yielded four OLS coefficients per model × preference combination.

We then estimated a linear mixed-effects regression using these OLS coefficients as the dependent variable and a set of dummy variables indicating the regression specification (baseline versus demographics versus country versus demographics and country) as predictors. Preference-domain dummies (for the six preference types) were included as additional controls to account for systematic differences across domains. To capture unobserved heterogeneity across LLMs, we included random intercepts at the model level. The linear mixed-effects regression was performed using the statsmodels Python package (v.0.14.5)[100].

Prediction-error decomposition

To attribute individual-level prediction-error variation to observable factors, we first constructed, for each LLM, an averaged prediction-error measure by averaging the individual-level MSEs across the six preferences. We then ran eight OLS regressions of this MSE on each of the following predictors: the three demographic variables (gender, age and cognitive ability), the four country-level characteristics (log GDP per capita, IDI, log population and log internet users) and country fixed effects. In addition, we ran a regression that included all eight predictors jointly.

Ablation and other robustness experiments

We designed four types of personas (six conditions in total) in the ablation experiments: full-persona (GPT-5.4-mini in the main text, with all demographic and country information included in system prompts), demographic-persona (demographic information only), country-persona (country information only) and partial-persona (full-persona with one of age, gender or cognitive ability removed). All ablation experiments were conducted in local languages, consistent with our original experiments (see Supplementary Information section F for detailed templates).

In the paraphrasing experiment, each GPS item was rewritten to preserve the meaning while altering the surface wording. In the response-constraint experiment, we restricted the answer format by instructing the LLM to respond only with a single number (Supplementary Information section G). In the high-temperature experiment, we reran GPT-5.4-mini with the temperature increased from 0 to 1 to test whether the temperature setting in the main analyses understates response diversity (Supplementary Information section J).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data used in this paper are available via the Open Science Framework at https://osf.io/pt2n4. The GPS data are publicly available at https://gps.econ.uni-bonn.de/home. All country-level auxiliary data are from publicly available sources, as documented in Supplementary Information section K. Source data are provided with this paper.

Code availability

The code for the analyses presented in this Article is available via the Open Science Framework at https://osf.io/pt2n4.

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Author contributions

F.C., Q.W. and S.B. designed the research. F.C., L.W. and S.B. supervised the research. F.C. and Q.W. performed the research. F.C. and Q.W. analysed the data. F.C., Q.W. and S.B. wrote the paper.

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Competing interests

The authors declare no competing interests.

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Software and code

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Data collection	We analyzed secondary data from the Global Preferences Survey. We used Python scripts to collect LLM-generated data via the official APIs of OpenAI, Google, and DeepSeek. Questionnaires and prompts were translated into local languages using the Google Translate API (and the GPT-4o-mini API for Dari, which Google Translate does not support).
Data analysis	We used Python (version 3.12.4) and Stata 18MP for data analysis. The following Python packages were used: NumPy (1.26.4), pandas (2.2.2), SciPy (1.16.1), Matplotlib (3.9.2), seaborn (0.13.2), Pingouin (0.5.5), statsmodels (0.14.5), and tqdm (4.66.5). In Stata, we used the norm package. All code is available in our repository (https://osf.io/pt2n4).

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We used the Global Preference Survey (GPS) dataset as the human preference benchmark. The GPS dataset is publicly available at https://gps.econ.uni-bonn.de/home. All LLM-generated data and the processed GPS data used in our analyses are available in our repository (https://osf.io/pt2n4/). GDP, population, the number of internet users, the ICT Development Index, gender-related indices, and cultural-dimension indices are publicly available. Links to access these data are provided in the Supplementary Information.

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Reporting on sex and gender	We used the binary gender variable contained in the GPS dataset, which serves as our benchmark. Gender information was extracted from the GPS dataset and embedded in our system prompts. We compared gender differences in preferences estimated from the LLM-generated data with those from the human survey data (i.e., the GPS dataset).
Reporting on race, ethnicity, or other socially relevant groupings	We did not use race or ethnicity data. We extracted demographic, country and region variables from the GPS dataset and embedded them into our system prompts. The definitions of all these variables are consistent with the GPS program.
Population characteristics	We used the GPS as our human benchmark and used LLMs to generate simulated preference responses for each GPS respondent, constructing a virtual GPS. The analysis sample comprises 67,201 matched respondents from 74 countries. Of these, 54.7% identified as female and 45.3% as male; the average age was 42.45 years (s.d. = 17.56 years).
Recruitment	We did not recruit any human participants for this study. All human data are drawn from the existing, publicly available Global Preference Survey.
Ethics oversight	The Ethics Committee of the Neuromanagement Laboratory, Zhejiang University, determined that this study, which involves only the analysis of existing secondary data, was exempt from ethics review.

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Study description	This study used 20 LLM configurations (spanning models from OpenAI, Google, and DeepSeek) to generate simulated responses to economic preference survey items, with the Global Preference Survey (GPS) as the human benchmark. We assessed the alignment between LLM-generated data and the human benchmark data at the individual, demographic, country, and cultural levels. All data are quantitative.
Research sample	Our human data come from the Global Preference Survey (GPS) dataset, available at https://gps.econ.uni-bonn.de/home. The GPS elicits six experimentally validated preference measures (risk-taking, patience, positive reciprocity, negative reciprocity, altruism, and trust) from about 80,000 respondents in 76 countries. The GPS sample is representative by design. Based on the GPS dataset, we used LLMs to generate simulated preference responses for each respondent in a one-to-one manner; Supplementary Information Section A provides detailed sample information for each model and country. The GPS was chosen because it is the most comprehensive, experimentally validated, and globally representative dataset of economic preferences.
Sampling strategy	Sampling was conducted by the Global Preference Survey project. The GPS data were collected within the framework of the 2012 Gallup World Poll, and participants were selected through probability sampling designed to be nationally representative. No a priori sample size calculation was performed; we used the full available GPS sample, and the LLM-simulated sample was determined by the one-to-one matched design (one simulated response set per retained respondent per model).

Data collection	We did not collect any new human data; the human benchmark is the existing GPS dataset. The LLM-generated data were collected programmatically via the official APIs of OpenAI, Google, and DeepSeek, using Python scripts that submitted one survey item per API request, with respondent personas encoded in system prompts.
Timing	We collected GPT-4o-mini data from August 7 to August 9, 2024, Gemini 2.0 Flash and DeepSeek-V3 data from March 7 to March 19, 2025, GPT-3.5-turbo, GPT-4.1-mini, GPT-4o-mini (English) data from April 28 to May 6, 2025, GPT-4o data from September 12 to September 14, 2025, and Gemini 2.5 Flash data from September 14 to October 4, 2025. We collected GPT-5.4 and GPT-5.4-mini data (including the robustness experiments: ablations, paraphrasing, and response-constraint experiments) from March 23 to April 9, 2026, and GPT-5.4-mini (high-temperature experiment) data from May 8 to May 9, 2026.
Data exclusions	Morocco was excluded because the translation API does not support Moroccan Arabic; Cambodia was excluded because pilot data showed a high rate of non-parsable or off-scale responses. This yields 74 of the original 76 countries. Within each country, only respondents surveyed in the country's most widely sampled language were simulated. Respondents were retained in the merged analysis sample only if valid responses were obtained from all twelve LLM configurations, yielding a final sample of 67,201 respondents (approximately 84% of the original GPS sample).
Non-participation	No new human participants were recruited, so no participants dropped out or declined participation. For the LLM simulations, Morocco was not included because the translation API does not support Moroccan Arabic, and Cambodia was not included owing to a high rate of non-parsable or off-scale responses in pilot data.
Randomization	Randomization is not applicable: this is an observational audit comparing LLM-simulated responses with existing survey data, with no allocation of participants into experimental groups. Every retained GPS respondent was simulated under all LLM configurations.

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☒	☐ Plants

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